

Curiosity Weekends
A Community Centre for Side Projects

Operations Report

April - May 2025

Dated: 2025-06-01
someone@curiosityweekends.org
+91 9349101566

What is Curiosity Weekends?

Curiosity Weekends is a community-driven tinkering initiative, primarily led by high school students, focusing on:

- Learning through making and tinkering
- Peer-to-peer learning

Genesis

In mid-2024, a group of students began gathering on the second Saturday of each month, emphasizing:

- Learning through sharing
- Puzzle-solving and treasure hunts
- Hackathons

By March 2025, the need for a shared workspace during summer vacation led to discussions with Linnk Academy. This collaboration resulted in the establishment of the Curiosity Lab on the first floor of ASAP, Vidyanagar, Kasaragod, generously provided by Linnk Academy.

Philosophy

A baby learns to walk not by studying physics but through curious, playful attempts and learning from failures. Similarly, learning, skill-building, product development, and even running a factory can be enjoyable when approached with curiosity and a willingness to embrace failures sustainably.

The Last Two Months: April and May 2025

In the past two months, we established the Curiosity Lab with the following facilities:

- A working facility with a capacity of 14 seats
- Spare laptops for use
- An electronics development station with tools and spare parts
- NodeMCUs, Arduinos, and Raspberry Pis
- A soldering station
- A prototyping station with breadboards and multimeters
- 3D, Virtual, and Augmented Reality Centers
 - 3D printer
 - 3D pen
 - Meta Quest VR/AR corner
 - Microsoft Kinect 3D cameras
- Tools: electric cutters, electric drills, and hand drills

Supporters

- Linnk Academy
- Little KITES, Government of Kerala
- Technical High School Kasaragod: Provided their workshop for the Carpentry Hackathon

Mentors

- Roji Joseph, KITE, Kasaragod
- Dr. Jijo, Professor,, Department of Physics, Government College, Kasaragod
- Jayaprakash Shetty DB, Drawing Teacher, SATHSS Manjeshwar
- Jemshid KK, Applied Mathematician

Events and Programs

- **Aeromodelling Workshop:** Experimented with paper planes to achieve the longest endurance
- **Carpentry Hackathon:** Built a wooden shelf for tools using the carpentry workshop at Technical High School
- **Ente Humanoid:** Showcased a humanoid at the “Ente Keralam” expo, a major attraction
- Managed the Education Department stall for “Ente Keralam”

Side Projects in Progress

- **TemPro: Temperature Profiler**
 - **Issue:** Uneven temperature distribution in the Curiosity Lab led to complaints
 - **Solution:** 10 IoT temperature sensors integrated into a web dashboard via MQTT, with calibration for offset error correction
 - **View at:** curiosityweekends.org/tempro
 - **Next:** A portable version in development
- **Humanoid**
 - Built a 1:1 scale humanoid from scratch
 - **Features:** AI voice interaction, eyes synced with speech, head tracks people, 1-degree-of-freedom hands
 - Version 1.0 released
- **GetMyBus**
 - **Motivation:** Lab attendees travel by bus and need better tracking
 - An IoT project with browser- and app-based frontends for bus arrival estimates
- **RFID Attendance Tracker**
- **Autonomous Robo Car**
- **Asset Management Software for the Curiosity Lab**
- **Clock-In**
 - A check-in/out and task tracker for the lab
- **Steady Hands**
 - A fun wire-tracking game to test hand control
- **Thermocol Cutter**
 - A mini portable cutter using Nichrome resistance wire

Achievers from the Community

- **Sivanand:** Won second prize in a statewide hackathon by Styep
- **Sreenand:** Delivered a young entrepreneurs session at TALROP, Village Park, Kasaragod

Roadmap

Over the past two months, during summer vacation, the lab hosted approximately 20 people daily, both in-person and on Discord. As school resumes, we will shift to a weekends-focused mode, continuing side projects and seeking broader collaborations with:

- Prominent core technology companies in India
- Government College, Kasaragod
- Central University of Kerala, Kasaragod

Asks and Gives

The community operates without formal fundraising, relying on voluntary contributions of components and equipment from individuals and organizations. We plan to continue this model. Below are specific requests from student groups, paired with their commitments, with detailed proposals available on our Git repository and community website. All contributions and deliverables are executed peer-to-peer, with updates shared with the Curiosity Weekends community.

- **Electronic Workbench**
 - **Ask:** Oscilloscope, function generator, lab power supply, temperature-controlled soldering station with magnifier (Cost: INR 75,000)
 - **Give:** A well-maintained electronics development center for the broader community to experiment and learn
- **Temperature Profiler**
 - **Ask:** Electronic components (Cost: INR 10,900)
 - **Give:** A fully functional IoT-based temperature monitoring system, scalable and portable sensor node architecture, real-time browser-based dashboard, and efficient MQTT data communication
- **Line-Following and Obstacle-Avoidance Robot**
 - **Ask:** Electronic components (Cost: INR 2,790)
 - **Give:** A working car with design and code
- **Bluetooth-Controlled Car**
 - **Ask:** Electronic components (Cost: INR 2,350)
 - **Give:** A Bluetooth-controlled car with design and code

Attendance

SI #	Name	Designation	Organization	What we did	What next
1	Ishan Jemshid	Student, 10th Std	CHSS Chattanchal		
2	Sinan	Student, 10th Std	THS Kasaragod	Humanoid robot, Carpentry Hackathon, Temprow, Mentor at Robotics hackthon	Starting a Startup called Nextbot labs, want to start working on paid projects
3	Hadin	Student, 11th STD	CJHSS Chemnad		
4	Sivanand	Student, 10th STD			
5	Mehran	Student, 11th STD	GVHSS Mogral		
6	Harsh	Student, 12th STD	Canara Pre-University College		
7	Sreenand	Student, 12th STD	GVHSS Iriyanni		
8	Arshad	Student, 10th STD	THS Kasaragod		
9	Sahad	Student, 10th STD	THS Kasaragod		
10	Sidhique	Student, 9th Std	THS Kasaragod		
11	Althaf	Student, 10th STD	Thabeeth HSS, Vidyanagar		
12	Ahmed	Student, 11th STD	MP International School		
13	Jacob	System Engineer	Infosys		
14	Jemshid	Applied Mathematician	Timesmix Interactive Pvt. Ltd.	Carpentry hackthon	Refine humanoid look at feature
15	Afkar	Student, 11th STD	GVHSS Mogral		Starting a startup called ElectroSparkzz
16	Salman	Mern stack developer	Brototype		
17	Roji Joseph	District Co-Ordinator	Kite Kerala		